

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (A)

RVS NAGAR, CHITTOOR

III B.Tech-I Semester: Sessional Examination-II

Subject name: – Natural Language Processing and Models Subject code: 23ACM01 Time: 2 Hours

Branch :CSM & CAI

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 =10 M

1. Write any two applications of RNNs in NLP.
2. How activation functions are used in the neural network process.
3. How to Pre-training and Fine-tuning in BERT.
4. Define Masked Language Modeling (MLM).
5. Define Hybrid machine translation.

Part B

Answer all the questions, each question carry equal marks

3 x 5 =15 M

- 6.(a) Write in detail about Named entity recognition (NER) using bidirectional long short-term memory (Bi-LSTM) in NLP. How Bi-LSTMs work for NER?

OR

- 6.(b) How RNNs process sequential data? Explain with example.

- 7.(a) Explain the working mechanism of Encoder-Decoder Model.

OR

- 7.(b) Describe the fine-tuning process in text classification with transformers.

- 8.(a) What is Neural machine translation (NMT)? Describe with example.

OR

- 8.(b) Discuss about Hybrid machine translation (HMT) with its different approaches.

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III B.Tech-I Semester: Sessional Examination-II

Subject name: – Operating Systems & System Programming

Subject code: 23ACA06

Time: 2 Hours

Branch : CSM & CAI

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

1. What is the role of a device controller in I/O systems?
2. Define Direct Memory Access (DMA).
3. What is an address space? How is it relevant to system software?
4. Differentiate between system programming and application programming.
5. Mention any two types of memory allocation during compilation.

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

- 6.(a) Discuss the security design principles in operating systems with examples.

OR

- 6.(b) What is user authentication? Explain different user authentication mechanisms.

- 7.(a) Explain the software hierarchy suitable with examples.

OR

- 7.(b) Discuss advanced features of design of macro assemblers different from ordinary assemblers?

- 8.(a) Discuss about the concept of the semantic gap in compilation.

OR

- 8.(b) Explain about the pipes, signals, and threads in Unix-based system programming

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III B.Tech-I Semester:

Sessional Examination-II

Subject name: – Computer Vision & Image Processing

Subject code: 23ACM02

Branch :CSM & CAI

Time: 2 Hours

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 =10 M

1. Define epipolar geometry in stereo vision.
2. What is disparity mapping?
3. Mention one feature matching algorithm.
4. What is template matching in object recognition?
5. Differentiate between lossy and lossless image compression.

Part B

Answer all the questions, each question carry equal marks

3 x 5 =15 M

- 6.(a) Describe disparity mapping techniques and their role in depth estimation.

OR

- 6.(b) Explain epipolar geometry in stereo vision with neat diagrams.

- 7.(a) Explain the Scale-Invariant Feature Transform (SIFT) algorithm with its main steps.

OR

- 7.(b) Discuss Speeded-Up Robust Features (SURF) and compare it with SIFT.

- 8.(a) Explain dilation and erosion in morphological image processing with neat diagrams.

OR

- 8.(b) Describe opening and closing operations in morphological processing. Compare their uses.

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III B.Tech-I Semester:

Sessional Examination-II

Subject name: – Data Analysis with Python

Subject code: 23ACD11

Time: 2 Hours

Branch :CSM & CAI

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 =10 M

1. Define central tendency.
2. What is a histogram?
3. Name two libraries for data visualization in Python.
4. Define violin plot.
5. Define feature engineering.

Part B

Answer all the questions, each question carry equal marks

3 x 5 =15 M

- 6.(a) Explain measures of central tendency and dispersion with examples.
OR
- 6.(b) Compare Histogram, Boxplot, KDE, and Bar Chart.
- 7.(a) Explain data visualization using Matplotlib with examples.
OR
- 7.(b) Discuss violin plots, strip plots, and swarm plots with use cases.
- 8.(a) | Perform step-by-step EDA on Titanic dataset (handling missing values, visualization, insights).
OR
- 8.(b) Explain EDA on Iris dataset with univariate and bivariate analysis.

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III B.Tech-I Semester: Sessional Examination-II

Subject name: – Introduction to Quantum Technologies and Applications

Subject code: 23ACS27

Branch : Common to all Branches

Time: 2 Hours

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

1. Name two hardware platforms used to build quantum computers.
2. Which hardware platform uses ions trapped in electromagnetic fields?
3. What is Quantum Key Distribution (QKD)?
4. Give one difference between classical gates and quantum gates.
5. Name one emerging career role in the quantum field.

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

6. (a) Explain conceptually what is required to build a quantum computer.

OR

6. (b) Why is maintaining entanglement difficult? Discuss the theoretical barriers.

7. (a) Describe the working of Quantum Key Distribution (QKD) and its importance in secure communication.

OR

7. (b) Discuss the real-world importance and future potential of quantum technologies in computing and communication.

8. (a) Discuss the role of quantum sensing and precision timing in real-world scenarios.

OR

8. (b) Discuss the educational and research opportunities for India in the global quantum race.